Day 1 – Java Basics + Time Complexity (Detailed)

1️⃣ DSA kya hota hai?

DSA = Data Structures + Algorithms

- **Data Structure** → data ko store aur organize karne ka tarika
- **Algorithm** → problem solve karne ka step-by-step method

Example:

Agar tumhe **numbers search** karna hai:

Array → Data Structure
Binary Search → Algorithm

2️⃣ Java Basics Revision

DSA start karne se pehle ye 5 Java topics clear hone chahiye.

1. Variables

Variable ek container hota hai jo value store karta hai.

Example:

```
int a = 10;
int b = 20;

System.out.println(a + b);
```

Explanation

- int → data type
- a → variable name
- 10 → value

Output

```
30
```

2. Data Types

Java me basic data types:

Data Type	Example
int	10
double	10.5
char	'A'
boolean	true / false
String	"Hello"

Example:1.

```
int age = 21;
double price = 99.5;
char grade = 'A';
boolean isStudent = true;
```

3. Loops

Loop ka use **repeat kaam** ke liye hota hai.

For Loop

```
for(int i = 1; i <= 5; i++){
    System.out.println(i);
}
```

Output

```
1
2
3
4
5
```

Explanation:

- Start → i = 1
- Condition → i <= 5
- Increment → i++

While Loop

```
int i = 1;

while(i <= 5){
    System.out.println(i);
    i++;
}
```

3️⃣ Functions (Methods)

Function ek reusable code block hota hai.

Example:

```
public class Main {

  static int add(int a, int b){
      return a + b;
  }

  public static void main(String[] args){
      int result = add(5, 3);
      System.out.println(result);
  }
}
```

Output

8

4️⃣ Arrays (Important for DSA)

Array ek **same data type elements ka collection** hota hai.

Example:

```java
int[] arr = {10,20,30,40};

System.out.println(arr[0]);
```

Output

10

Important Points:

- Array index **0 se start hota hai**
- arr.length → array size

Array traversal example:

```java
int[] arr = {10,20,30,40};

for(int i = 0; i < arr.length; i++){
    System.out.println(arr[i]);
}
```

Output

10
20
30
40

5️⃣ Time Complexity (Very Important)

Time complexity batata hai **algorithm kitna fast ya slow hai**.

Hum use **Big O Notation** se likhte hain.

Common complexities:

Complexity	Meaning
O(1)	Constant
O(n)	Linear
O(log n)	Logarithmic
$O(n^2)$	Quadratic

Example 1 – O(1)

int a = 10;
System.out.println(a);

Input kuch bhi ho, time same.

👉 **O(1)**

Example 2 – O(n)

```
for(int i = 0; i < n; i++){
    System.out.println(i);
}
```

Agar n = 100

Loop = 100 times

👉 **O(n)**

Example 3 – $O(n^2)$

```
for(int i = 0; i < n; i++){
    for(int j = 0; j < n; j++){
        System.out.println(i + " " + j);
    }
}
```

Agar n = 10

Total operations = 10 × 10

👉 **O(n^2)**

6️⃣ Space Complexity

Space complexity batata hai **program kitni memory use karta hai**.

Example:

```
int a = 10;
```

Memory use → constant

👉 **O(1)**

7️⃣ Day 1 Practice Problems

Aaj ye programs Java me likho.

Problem 1 – Print numbers

```
for(int i = 1; i <= 10; i++){
    System.out.println(i);
}
```

Problem 2 – Sum of numbers

```
int sum = 0;

for(int i = 1; i <= 10; i++){
    sum = sum + i;
}
```

System.out.println(sum);

Output

55

Problem 3 – Find maximum in array

```
int[] arr = {5,9,2,7};

int max = arr[0];

for(int i = 1; i < arr.length; i++){
```

```
    if(arr[i] > max){
        max = arr[i];
    }
}

System.out.println(max);
```

Output

```
9
```